Problem

Stewart, *Calculus: Early Transcendentals*, Section 15.7 (Triple Integrals), Exercise 28.

Sketch the solid whose volume is given by the iterated integral

$$\int_0^2 \int_0^{2-y} \int_0^{4-y^2} dx\, dz\, dy.$$

The solution must identify the bounding surfaces, draw the solid, and compute the volume as a check.

Part (a)

Notation

The symbol E is the solid region in space whose volume the integral measures. The symbol dV is the volume element. For the order $dx\, dz\, dy$ it is

$$dV = dx\, dz\, dy.$$

An *iterated integral* evaluates $\iiint_E f\, dV$ as three nested single integrals. Fubini's theorem says that the value does not depend on the order when f is continuous on E and every order that fits the geometry is used. The letter y names the vertical coordinate in this problem only because the limits are written that way; all statements below refer to the coordinate axes as drawn in Figure 1.

Step 1: read the three pairs of limits

Read the integral from the innermost differential outward.

- Innermost, dx: $0 \le x \le 4 - y^2$. For fixed y and z the variable x runs from the plane $x = 0$ to the curved surface $x = 4 - y^2$.
- Middle, dz: $0 \le z \le 2 - y$. The variable z runs from the plane $z = 0$ to the plane $y + z = 2$.
- Outermost, dy: $0 \le y \le 2$.

The surface $x = 4 - y^2$ contains no z. A surface with an equation that is free of one coordinate is a *cylinder* whose rulings are parallel to the axis of that coordinate. Therefore $x = 4 - y^2$ is a parabolic cylinder with vertical rulings parallel to the z-axis. Its trace in every plane $z =$ constant is the same parabola, which opens toward negative x from the vertex $(4, 0)$ in the plane $z = 0$.

Step 2: the solid described by the three inequalities

The three pairs of limits are equivalent to

$$E = \{(x, y, z) : 0 \le y \le 2,\ 0 \le z \le 2 - y,\ 0 \le x \le 4 - y^2\}.$$

The variables y and z alone obey $0 \le y \le 2$ and $0 \le z \le 2 - y$. This pair of conditions describes the triangle

$$T = \{(y, z) : y \ge 0,\ z \ge 0,\ y + z \le 2\}$$

with vertices $(0, 0)$, $(2, 0)$ and $(0, 2)$ in the yz-plane. The triangle T is the shadow of E on the plane $x = 0$. Above each point of T the solid runs from $x = 0$ to the parabolic cylinder $x = 4 - y^2$.

At $y = 2$ the cylinder has $x = 0$ and the plane $y + z = 2$ has $z = 0$. The two surfaces meet at the single point $(0, 2, 0)$. The solid therefore tapers to a point at that corner.

Step 3: the five faces and the vertices

Table 1 lists the five faces. Two of them are plane faces of the triangle T, two are the planes $x = 0$ and $y = 0$, and one is the curved face of the parabolic cylinder.

Table 1: The five faces of the solid E of Exercise 28.

Face	Surface	Vertices	Area
back	$y = 0$	$(0, 0, 0)$, $(4, 0, 0)$, $(4, 0, 2)$, $(0, 0, 2)$	8
side	$x = 0$	$(0, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$	2
floor	$z = 0$	$(0, 0, 0)$, $(4, 0, 0)$, $(0, 2, 0)$	$16/3$
top	$y + z = 2$	$(4, 0, 2)$, $(0, 0, 2)$, $(0, 2, 0)$	$16\sqrt{2}/3$
curved	$x = 4 - y^2$	$(4, 0, 0)$, $(4, 0, 2)$, $(0, 2, 0)$	3.5358

The five faces use the surfaces

$$y = 0, \qquad x = 0, \qquad z = 0, \qquad y + z = 2, \qquad x = 4 - y^2.$$

The solid is the set of points that lie on the inner side of all five surfaces. The test point $(1, 0.5, 0.5)$ gives $x = 1 < 4 - 0.25 = 3.75$ and $z = 0.5 < 1.5$, so the test point is inside E.

The solid has five vertices:

$$(0, 0, 0), \quad (4, 0, 0), \quad (4, 0, 2), \quad (0, 0, 2), \quad (0, 2, 0).$$

The point $(0, 2, 0)$ is the sharp corner where the curved face and the top face meet.

Step 4: the sections parallel to the plane $x = 0$

Let y be fixed with $0 \le y \le 2$. The points of E at that value of y satisfy

$$0 \le z \le 2 - y, \qquad 0 \le x \le 4 - y^2.$$

This set is a rectangle with side lengths $4 - y^2$ and $2 - y$. The area is

$$A(y) = (4 - y^2)(2 - y) = 8 - 4y - 2y^2 + y^3.$$

The area decreases from $A(0) = 8$ to $A(2) = 0$, because both side lengths decrease. The derivative is $A'(y) = -4 - 4y + 3y^2$, which is negative on $0 < y < 2$.

Figure 1 shows the solid. Figure 2 shows the shadow T and the graph of $A(y)$.

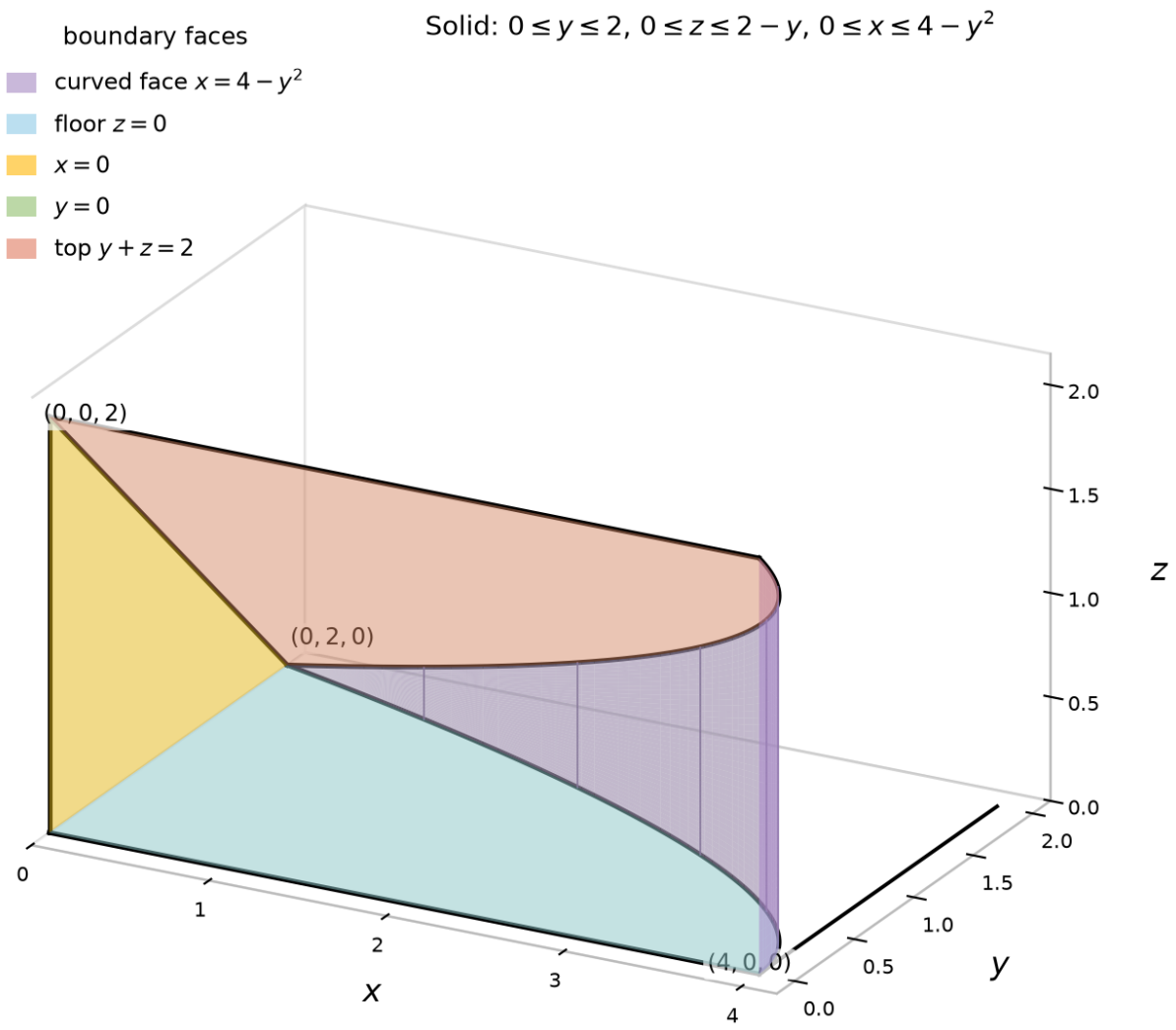


Figure 1: The solid of Exercise 28. The five boundary faces are the rectangle in the plane $y = 0$, the triangle in the plane $x = 0$, the floor $z = 0$ under the parabola, the plane top $y + z = 2$, and the curved face of the parabolic cylinder $x = 4 - y^2$. The thin vertical lines on the curved face are rulings of the cylinder; they are parallel to the z-axis.

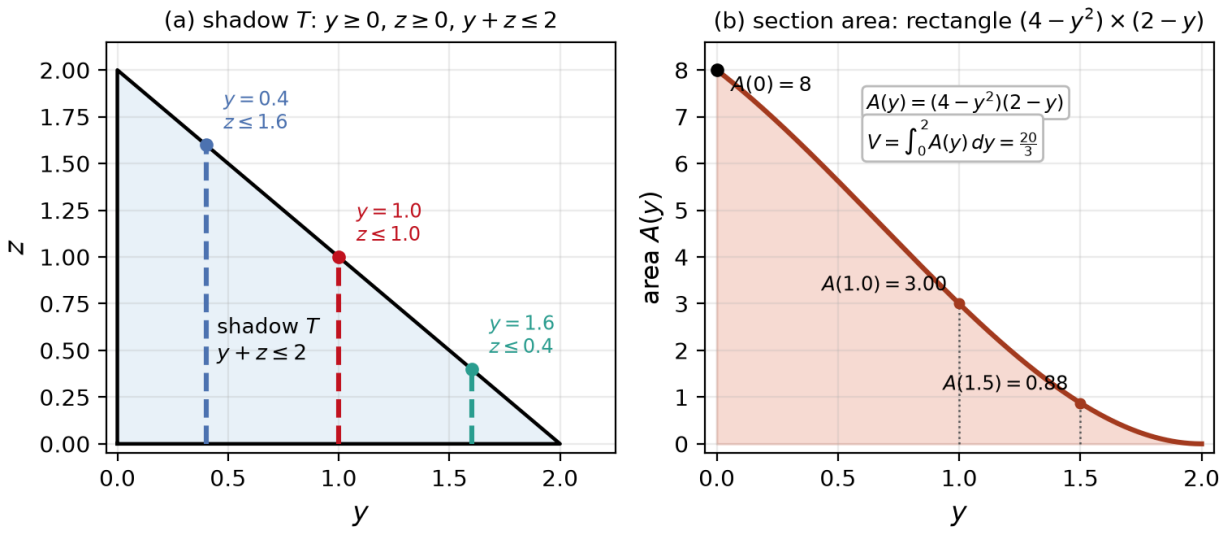


Figure 2: (a) The shadow T of the solid in the yz-plane is the triangle $y \geq 0$, $z \geq 0$, $y + z \leq 2$. A vertical line at y runs from $z = 0$ to $z = 2 - y$. (b) The area of the section at y is $A(y) = (4 - y^2)(2 - y)$. The shaded area under the graph is the volume $20/3$.

Step 5: the volume

The volume is the integral of the constant function 1 over E. Integrate first in x:

$$V = \int_0^2 \int_0^{2-y} \int_0^{4-y^2} 1 \, dx \, dz \, dy = \int_0^2 \int_0^{2-y} (4 - y^2) \, dz \, dy.$$

The integrand is free of z, so the middle integral multiplies the length $2 - y$:

$$V = \int_0^2 (4 - y^2)(2 - y) \, dy = \int_0^2 \left(8 - 4y - 2y^2 + y^3\right) dy.$$

The antiderivative is

$$8y - 2y^2 - \frac{2y^3}{3} + \frac{y^4}{4},$$

and the definite integral is

$$V = \left[8y - 2y^2 - \frac{2y^3}{3} + \frac{y^4}{4}\right]_0^2 = 16 - 8 - \frac{16}{3} + 4 = 12 - \frac{16}{3} = \frac{20}{3}.$$

Three independent checks follow.

Check 1, sections in z. Let z be fixed. Then $0 \leq y \leq 2 - z$ and $0 \leq x \leq 4 - y^2$, so

$$S(z) = \int_0^{2-z} (4 - y^2) \, dy = \left[4y - \frac{y^3}{3}\right]_0^{2-z} = 4(2 - z) - \frac{(2 - z)^3}{3}.$$

The outer integral gives

$$\int_0^2 S(z) \, dz = \int_0^2 \left(4(2 - z) - \frac{(2 - z)^3}{3}\right) dz = \frac{20}{3}.$$

Check 2, divergence theorem. For the vector field $\mathbf{r} = (x, y, z)$ the divergence is 3, so $3V$ equals the flux of $\mathbf{r}$ across the boundary. The faces $x = 0$, $y = 0$ and $z = 0$ contribute nothing, because $\mathbf{r} \cdot \mathbf{n} = 0$ on each of them. On the top face $y + z = 2$ the unit normal is $(0, 1, 1)/\sqrt{2}$ and $\mathbf{r} \cdot \mathbf{n} = 2/\sqrt{2} = \sqrt{2}$; with $dS = \sqrt{2}\, dx\, dy$ the flux is $2 \cdot (16/3) = 32/3$. On the curved face the normal is $(1, 2y, 0)/\sqrt{1 + 4y^2}$, so $\mathbf{r} \cdot \mathbf{n} = (4 + y^2)/\sqrt{1 + 4y^2}$ and $dS = \sqrt{1 + 4y^2}\, dz\, dy$. The flux is

$$\int_0^2 \int_0^{2-y} (4 + y^2)\, dz\, dy = \int_0^2 (4 + y^2)(2 - y)\, dy = \frac{28}{3}.$$

The total flux is $32/3 + 28/3 = 20$, hence $V = 20/3$.

Check 3, numerical. The verification scripts give the values in Table 2.

Table 2: Numerical values of the volume of Exercise 28.

Method	Value
SymPy, exact	$20/3 = 6.666666666667$
Composite Simpson rule, 200000 panels	6.666666666667
Gauss–Legendre, order $dx\, dz\, dy$, 140^3 nodes	6.666666666667
Gauss–Legendre, order $dz\, dx\, dy$, 130^3 nodes	6.666666666667
Monte Carlo, $4 \cdot 10^6$ points in the box $[0, 4] \times [0, 2] \times [0, 2]$	6.663780 ± 0.003944
Divergence theorem, fluxes 32/3 and 28/3	6.666666666667

$$\boxed{E = \{(x, y, z) : 0 \le y \le 2,\ 0 \le z \le 2 - y,\ 0 \le x \le 4 - y^2\}, \qquad V = \frac{20}{3}}$$

The solid is bounded by the planes $y = 0$, $x = 0$, $z = 0$, $y + z = 2$ and by the parabolic cylinder $x = 4 - y^2$.

How to solve problems of this kind

1. Read the iterated integral from the innermost differential outward and write one inequality for each pair of limits.
2. Identify the surface behind every inequality. A surface that misses one variable is a cylinder parallel to the axis of that variable; a surface that is linear in all variables is a plane.
3. Use the outer two limits to draw the shadow of the solid on a coordinate plane. In this exercise the shadow is a triangle, and the triangle fixes the outer limits of any order of integration.
4. Take sections parallel to the plane of the shadow. The section is often a rectangle, and its area is a product of two lengths. Integrate the area over the remaining variable.
5. Check the volume with a second method: sections in another direction, the divergence theorem, a geometric formula, or a numerical quadrature.
6. Watch for the points where two surfaces meet. They are the vertices of the solid, and they are the places where the limits of integration change.